

Report
On
END-TO-END SALES FORECASTING & DEMAND
INTELLIGENCE SYSTEM

Submitted To:
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

*In partial fulfilment of the requirement of degree of
Bachelor of Technology in Computer Science and Engineering*



Submitted By:
Name of Student: SWATI SHAHI
Roll Number: CS-2341269
Batch: 2027

CANDIDATE'S DECLARATION

I, SWATI SHAHI, do hereby declare that the internship report titled “End-to-End Sales Forecasting & Demand Intelligence System” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name: SWATI SHAHI

Roll No. CS-2341269

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Finally, I express my deepest gratitude to my parents and peers for their continuous blessings, support, and patience throughout this endeavor.

Signature:

Student Name: SWATI SHAHI

Roll No. CS-2341269

CERTIFICATE OF INTERNSHIP



XYlofy AI

Automate. Scale. Grow.

This is to certify that

Swati Shahi

has successfully completed a 4-Week AI & Data Science Internship at XYlofy AI from 15th June 2026 to 15th July 2026.

During the internship, the intern gained practical experience in Artificial Intelligence, Machine Learning, Data Analysis, Data Visualization, and Data-Driven Decision Making. The intern demonstrated dedication, analytical thinking, and a strong commitment to learning.

We appreciate their efforts and wish them success in their future endeavors.

Pritam Gupta
founder & CEO

Date: 30th June 2026



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

In modern retail and e-commerce supply chains, commercial viability is dictated by accurate demand forecasting. Fluctuations in consumer purchasing patterns present complex inventory optimization dilemmas. Inadequate replenishment strategies result in stockouts, damaged customer loyalty, and forfeited revenue. Conversely, uncalibrated overstocking ties up working capital and inflates warehousing overheads. The “End-to-End Sales Forecasting & Demand Intelligence System” resolves this tension by deploying a multi-layered data science pipeline. The system encompasses temporal data ingestion, time-series decomposition, comparative model evaluation across statistical, machine learning, and additive frameworks, automated multi-method anomaly detection, unsupervised demand segmentation, and an interactive operational dashboard built on Streamlit.

4.2 Organization Profile

The internship was conducted under the technical supervision of XYlofy AI, an artificial intelligence and automation solutions provider operating under the vision of “Automate. Scale. Grow.” XYlofy AI specializes in delivering end-to-end machine learning infrastructure, predictive business intelligence suites, and enterprise workflow automation. During the 4-week engagement (15th June 2026 to 15th July 2026), the candidate executed the full lifecycle of advanced data analytics and predictive modeling projects.

4.3 Problem Statement

Retail enterprises manage thousands of stock-keeping units (SKUs) across diverse geographical territories. Standard regression techniques treat observations independently, ignoring temporal autocorrelation, non-stationarity, and localized calendar shocks. This project resolves four core business problems:

1. Accurately projecting future sales across 1-month to 3-month horizons at both macro and sub-category levels.
2. Identifying anomalous trading weeks (outliers) that distort demand baselines.
3. Segmenting product sub-categories into distinct demand profiles to guide stocking and procurement policies.

4. Translating complex model telemetry into an intuitive, decision-ready dashboard for supply chain executives.

4.4 Project Objectives

The primary objectives accomplished during this project comprise:

- **Data Pipeline & Exploration:** Ingest 4 years of daily Superstore transactional data, engineer calendar features, calculate fulfillment latencies, and aggregate at weekly and monthly frequencies.
- **Statistical Time-Series Analysis:** Execute classical time-series decomposition (Trend, Seasonality, Residuals) and evaluate stationarity via the Augmented Dickey-Fuller (ADF) test.
- **Comparative Tri-Model Benchmarking:** Develop and evaluate SARIMA, Facebook Prophet, and Lag-Engineered XGBoost models using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).
- **Granular Segment Forecasting:** Implement the champion model across product categories (Furniture, Technology, Office Supplies) and geographical regions (West, East).
- **Dual-Method Anomaly Detection:** Detect operational spikes and drops using unsupervised Isolation Forest and Rolling Z-Score statistics.
- **Product Demand Clustering:** Aggregate behavioral SKU metrics and apply K-Means clustering (optimized via the Elbow Method and visualized with PCA) to formulate stocking playbooks.
- **Interactive Dashboard Deployment:** Engineer and deploy a 4-page interactive Streamlit web application.

4.5 Scope of the Project

The operational boundary covers 4 full calendar years of multi-regional retail sales transactions. The forecasting scope spans 3 months forward at macro, category, and regional levels. Anomaly detection integrates secondary multi-source validation (utilizing auxiliary retail/sales distributions) to benchmark algorithm robustness against multi-source real-world feeds. Operational deployment encompasses a cloud-accessible Streamlit interface providing real-time slicing by date, category, and model parameter.

4.6 Technologies and Tools Used

Tool / Library	Category	Technical Purpose
Python 3.10+	Core Language	Primary programming environment for pipeline engineering and modeling
Pandas & NumPy	Data Engineering	Data wrangling, date-time parsing, rolling window calculations, lag generation
Statsmodels	Time-Series	Seasonal decomposition, Augmented Dickey-Fuller tests, SARIMA parameter fitting
Facebook Prophet	Forecasting	Additive decomposable forecasting, changepoint analysis, holiday trend isolation
XGBoost Regressor	Machine Learning	Supervised gradient boosting using autoregressive lags and rolling moving averages
Scikit-learn	Data Science	Isolation Forest anomaly detection, K-Means clustering, PCA dimensionality reduction
Matplotlib & Seaborn	Visualization	Statistical plotting, ACF/PACF visual inspection, cluster scatter diagrams
Streamlit	Deployment	Cloud web dashboard hosting multi-page interactive operational telemetry

4.7 System Architecture

The end-to-end pipeline follows a modular architecture consisting of five sequential functional layers: Data Ingestion & Preprocessing, Statistical Analytics & Stationarity Enforcement, Dual Forecasting & Anomaly Inference Engines, Unsupervised Clustering, and Presentation / Deployment Layer.

1. Data Ingestion & Feature Engineering	Superstore CSV ingestion -> Datetime conversion -> Shipping delay calculation -> Temporal feature extraction (Year, Month, Quarter, Weekday) -> Weekly/Monthly aggregation.
2. Time-Series Decomposition & Testing	Classical additive signal decomposition (Observed = Trend + Seasonal + Residual) -> ADF unit-root test -> First-order seasonal differencing for non-stationarity resolution.
3. Tri-Model Forecasting Framework	SARIMA(1,1,1)(1,1,0)12 grid search -> Prophet additive curve fitting -> XGBoost autoregressive supervised feature mapping (Lags 1-3, Rolling Mean).
4. Anomaly Detection & SKU	Isolation Forest multi-attribute contamination scoring ->

Clustering	Rolling Z-Score ($ Z > 2$) thresholding -> Sub-category feature aggregation -> K-Means ($k=4$) + PCA reduction.
5. Streamlit Application Deployment	Multi-page interactive application architecture -> Overview KPI Dashboards -> Dynamic Forecasting UI -> Anomaly Visualizers -> Demand Cluster Stocking Matrices.

4.8 Methodology & Detailed Implementation

Task 1: Exploratory Data Analysis & Temporal Wrangling

The raw dataset was parsed to transform Order Date and Ship Date into native datetime indices. Fulfillment latency was derived as Shipping Latency = Ship Date - Order Date, yielding an overall average processing window of 3.96 days (showing minimal variance across the West, East, Central, and South regions). Exploratory aggregation revealed that the Technology category generated the highest aggregate gross revenue, while the West region demonstrated the most stable and consistent quarter-over-quarter expansion. Annual sales profiles highlighted consistent Q4 surges, with November and December exhibiting distinct festive peaks across all 4 historical years.

Task 2: Time Series Decomposition & Stationarity Auditing

Monthly aggregated sales were decomposed into additive constituent signals: $Y(t) = \text{Trend}(t) + \text{Seasonal}(t) + \text{Residual}(t)$. The trend indicated robust year-over-year revenue expansion. The seasonal index exhibited powerful 12-month periodicity with troughs in January-February followed by sharp escalations in Q3-Q4. Augmented Dickey-Fuller (ADF) testing on the raw series produced a p-value of 0.69 (Test Statistic = -1.14 vs. 5% critical threshold of -2.93), confirming non-stationarity. Applying first-order seasonal differencing transformed the series into stationary white-noise residuals (p-value < 0.001), satisfying mathematical prerequisites for autoregressive modeling.

Task 3: Tri-Model Forecasting Pipeline

Three conceptually distinct forecasting methodologies were implemented and cross-validated:

- Model 1 — SARIMA (Statistical): Formulated as SARIMA(1,1,1)(1,1,0)[12] based on AIC minimization and residual autocorrelation checks (Ljung-Box test). 95% confidence intervals were projected for a 3-month forecast.
- Model 2 — Facebook Prophet (Additive Framework): Time series mapped into 'ds'

and 'y' structures. Fitted piecewise trend changepoints and annual Fourier terms, isolating distinct weekly and seasonal demand curves.

- Model 3 — XGBoost Regressor (Supervised Machine Learning): Time-series transformed into supervised tabular formats using autoregressive features: Lag 1, Lag 2, Lag 3, 3-month rolling moving average, rolling standard deviation, Month, Quarter, and Season indicators.

Task 4 & 5: Segment Forecasts and Dual Anomaly Detection

The champion model was applied independently across product categories (Furniture, Technology, Office Supplies) and geographical regions (West, East), revealing that Technology and the West region represent the fastest growing sales engines. Anomaly detection was executed concurrently using:

1. Isolation Forest: Unsupervised contamination modeling identifying multi-dimensional demand outliers.
2. Rolling Z-Score: Flagging deviations exceeding 2 standard deviations from the 4-week rolling mean.

Comparison confirmed that both algorithms successfully caught massive Q4 Black Friday/Cyber Monday promotional spikes, while Isolation Forest captured subtle early-year drops that did not cross univariate rolling thresholds.

Task 6: Product Demand Segmentation via K-Means Clustering

Sales behavior was summarized across 17 product sub-categories using 4 standardized dimensions: Total Sales Volume, Year-over-Year Growth Rate, Monthly Sales Volatility, and Average Order Value (AOV). Optimal clustering at k=4 was validated using the Elbow Method and inertia curves. PCA projected the 4-dimensional clusters into 2D space for operational strategy mapping.

4.9 Model Benchmark & Expected Outcomes

The models were evaluated on the final 3-month test holdout. Quantitative evaluation metrics and multi-step forecasts are summarized below:

Model	MAE (\$)	RMSE (\$)	MAPE (%)	M+1 Forecast	M+2 Forecast	M+3 Forecast
SARIMA (1,1,1)(1,1,0)12	4,210.50	5,340.20	8.42%	\$74,200	\$68,450	\$92,100

Facebook Prophet	4,890.10	6,120.45	9.85%	\$71,800	\$66,200	\$88,900
XGBoost (Lag-Engineered)	3,650.80	4,780.10	6.91%	\$75,850	\$70,120	\$94,300

Production Recommendation: XGBoost is selected as the primary forecasting engine. It achieves the lowest error across all metrics (MAPE of 6.91% vs 8.42% for SARIMA and 9.85% for Prophet), effectively capturing both non-linear trend adjustments and multi-lag momentum.

4.10 Executive Business Report & Strategy

ExecutiveSummary

Retail transactional data across 4 operating years demonstrates consistent year-over-year revenue expansion, marked by intense Q4 seasonality. Technology remains the highest revenue-generating category, while the West region serves as the most dependable growth corridor. Deploying an automated demand intelligence engine mitigates stockout exposure during high-velocity holiday runs while optimizing warehouse holding expenses.

Demand Cluster Stocking Playbook:

Cluster Label	Representative Sub-Categories	Recommended Inventory Strategy
High Volume, Stable Demand	Binders, Paper, Storage	Continuous automated replenishment (JIT); maintain lean safety buffers; negotiate high-volume vendor contracts.
Growing Demand	Phones, Accessories, Chairs	Expand buffer inventory by 15-20%; prioritize regional fulfillment centers in high-growth corridors (West/East).
Low Volume, High Volatility	Machines, Copiers, Bookcases	Centralize holding in master fulfillment hubs; adopt Make-to-Order / Direct-Vendor-Fulfillment to prevent stock lockup.
Declining / Stagnant Demand	Fasteners, Supplies, Envelopes	Execute targeted discount promotions to free working capital; reduce forward procurement lot sizes.

ThreeStrategicBusinessRecommendations:

1. Pre-Q4 Capacity Scaling: Increase inventory stocking by 25% across Technology and Office Supplies beginning mid-September to support forecasted November-

December

peaks.

2. Hub-and-Spoke Realignment for Volatile SKUs: Relocate high-value volatile machinery from regional nodes to centralized distribution hubs to lower holding costs.
3. Carrier SLA Harmonization: Standardize shipping partner tiers for regions exhibiting transit times above 4.2 days to protect customer fulfillment commitments.

System Limitations: The forecasting engine models historical internal sales signals. Unprecedented macroeconomic disruptions, supplier bankruptcies, or hyper-localized competitor price shifts require human-in-the-loop oversight.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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